འབྲུག་རྒྱལ་འཛིན་གཙུག་ལག་སློབ་གྲྭ་ གཞི་།

College of Science and Technology
Rinchending: Bhutan

DBS101
Database Systems Fundamentals
(SS2026)

Practical 1 Report

Submitted By.
Name: Pelden Nidup
Student No.: 02250360
Program: BE 1SWE
Date: 13-3-2026

RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorized group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

Aim

To gain knowledge and practical understanding by exploring at least two different RDBMS software.

Theory

MySQL:

MySQL is a popular open-source Relational Database Management System (RDBMS) that is widely used in many web applications and software systems. It works using SQL, which allows users to create databases, store data, update information, and retrieve data whenever needed. MySQL supports different types of data such as numbers, text, and dates, making it flexible for storing and managing various kinds of information. It is also known for being reliable and scalable, meaning it can handle a large amount of data and many users at the same time. Because of its performance and ease of use, MySQL is commonly used in many websites and data-driven applications.

Oracle:

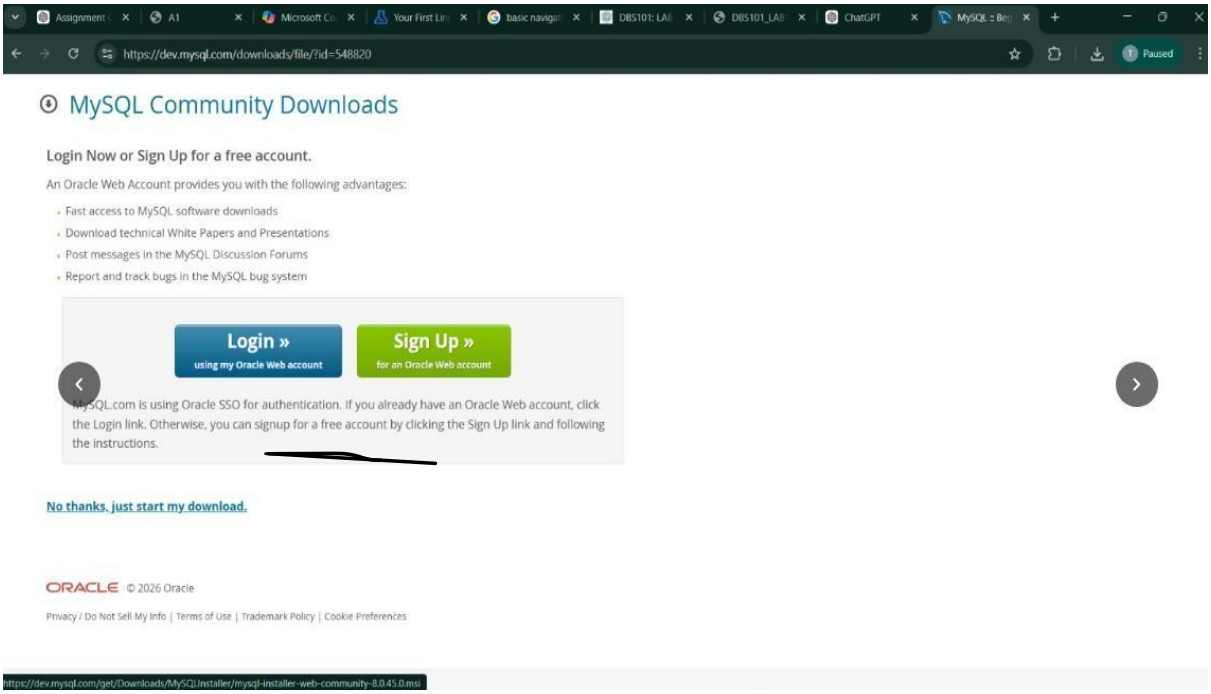
Oracle is another well-known Relational Database Management System developed by Oracle Corporation. It is mainly used by large organizations and enterprises because it offers many advanced features and strong performance. Like MySQL, Oracle also uses SQL to manage and retrieve data from databases. It is known for its strong security, high reliability, and ability to manage very large databases efficiently. Due to these features, many businesses, banks, and large institutions rely on Oracle to store and manage their important data and systems.

Implementation Steps

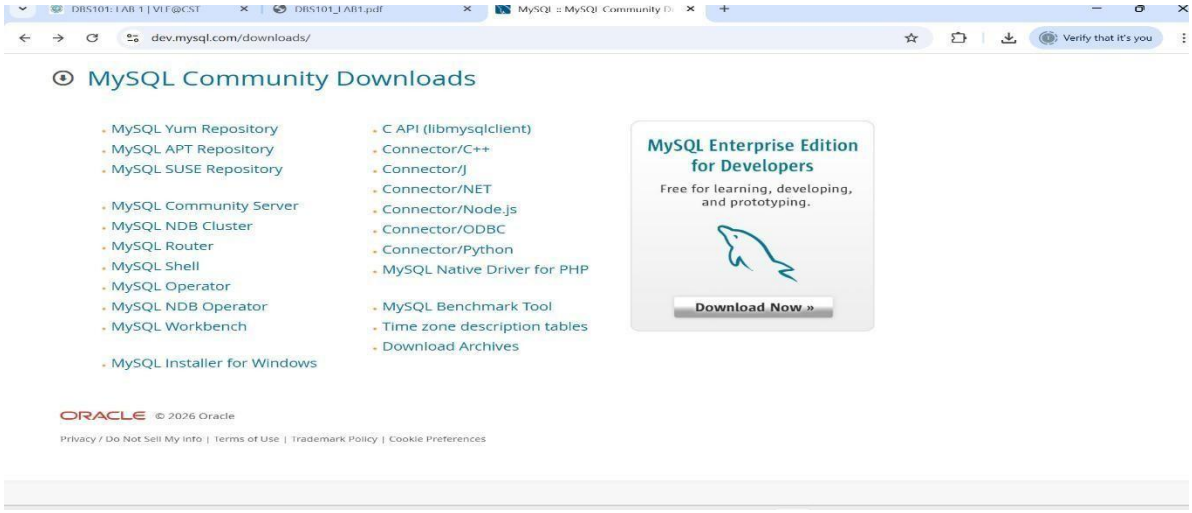
Work Evidence of Exercises; add screenshots of completed tasks

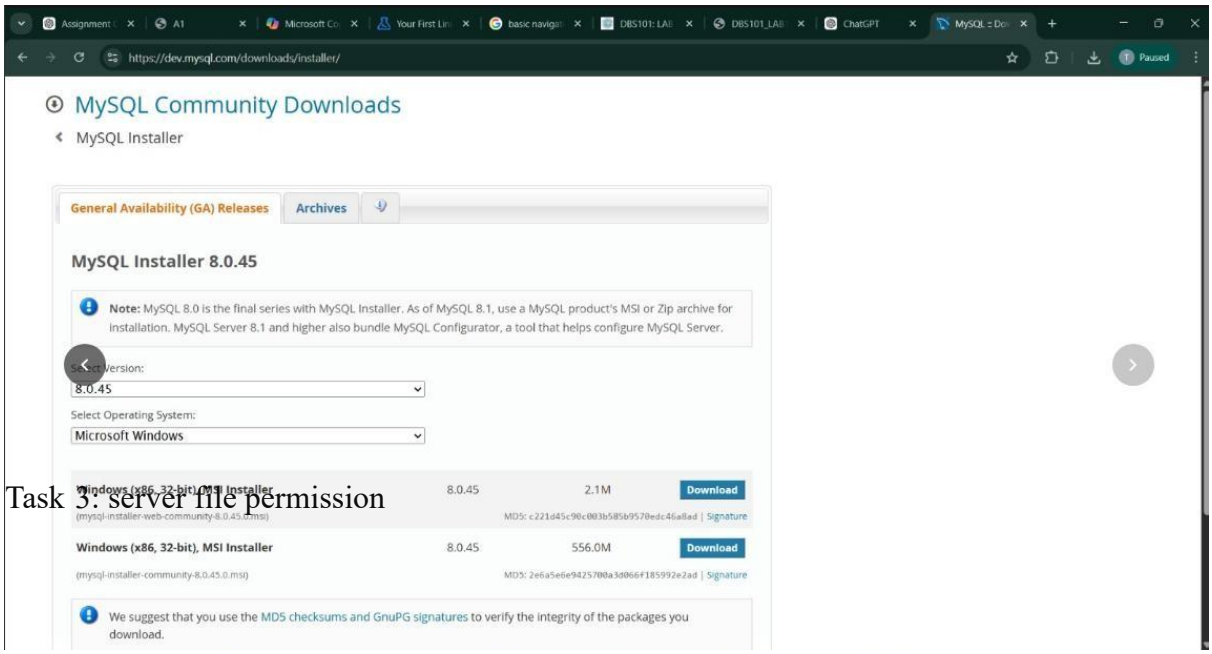
Example:

Task 1: MySQL Community Downloads



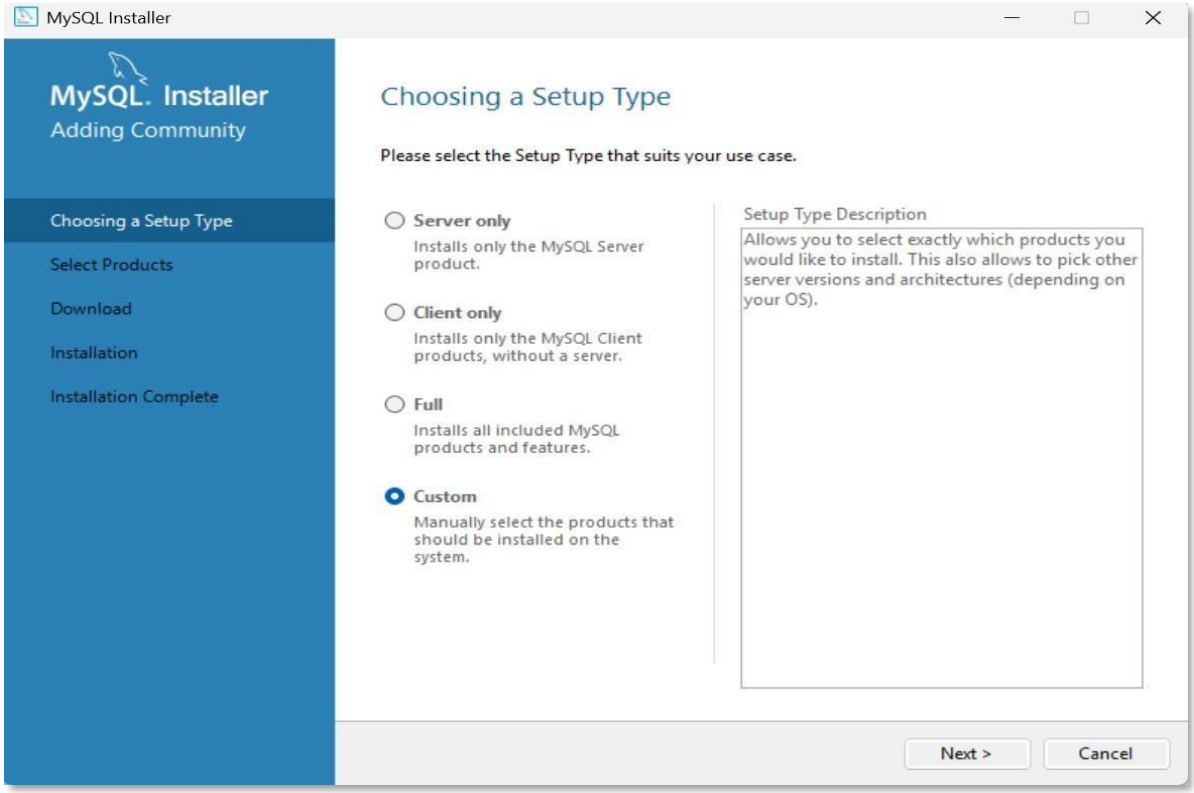
Task 2: Go to the MySQL Community Download page and choose the installer that matches your PC. Then, click on the first option: mysql-installer-web-community-8.0.45.0.msi. When prompted, select “No thanks, just start my download” to begin downloading it right away.



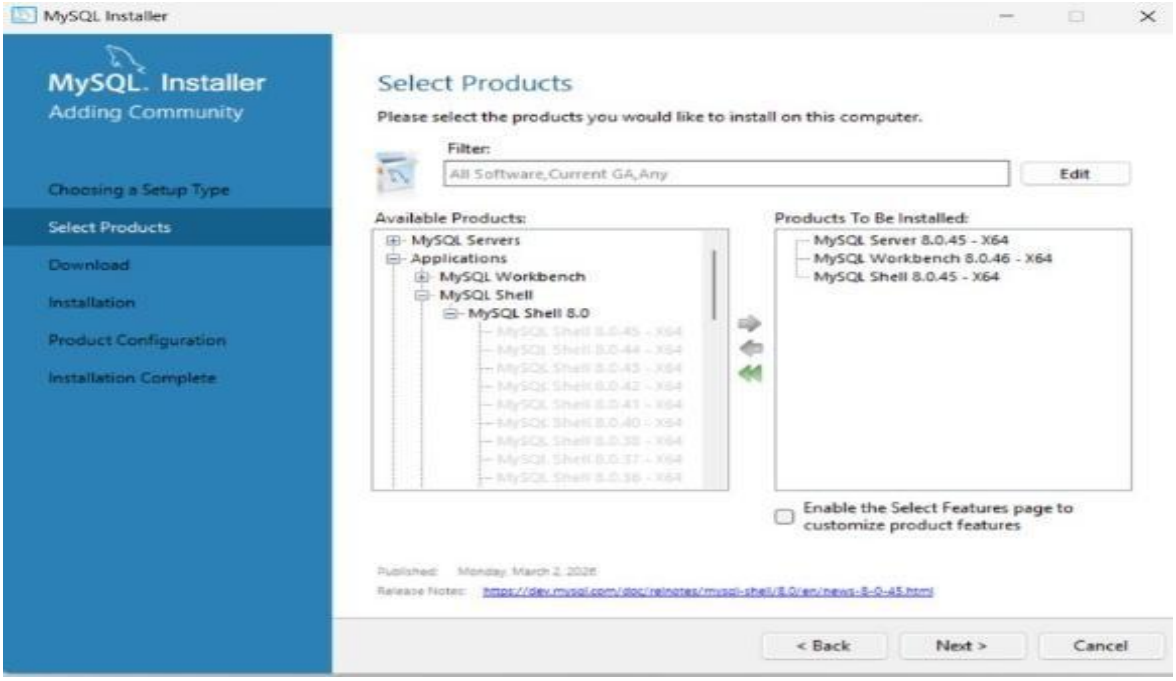


Task 3: server file permission

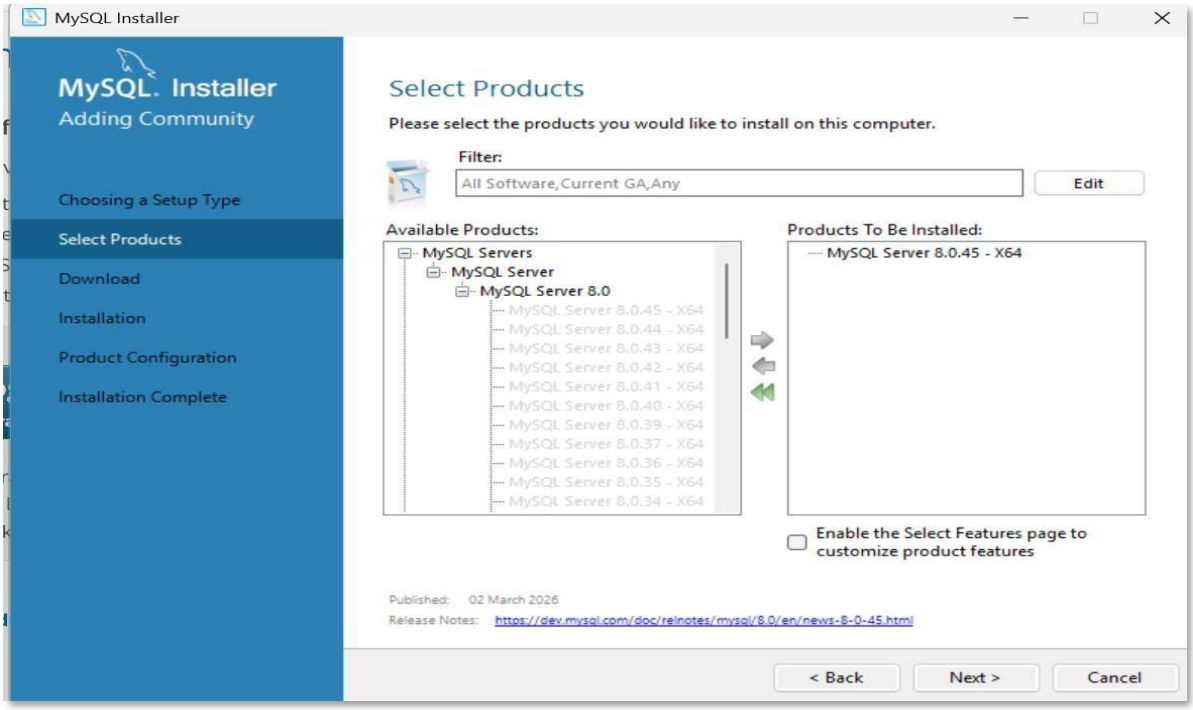
Task 3: Choose Custom



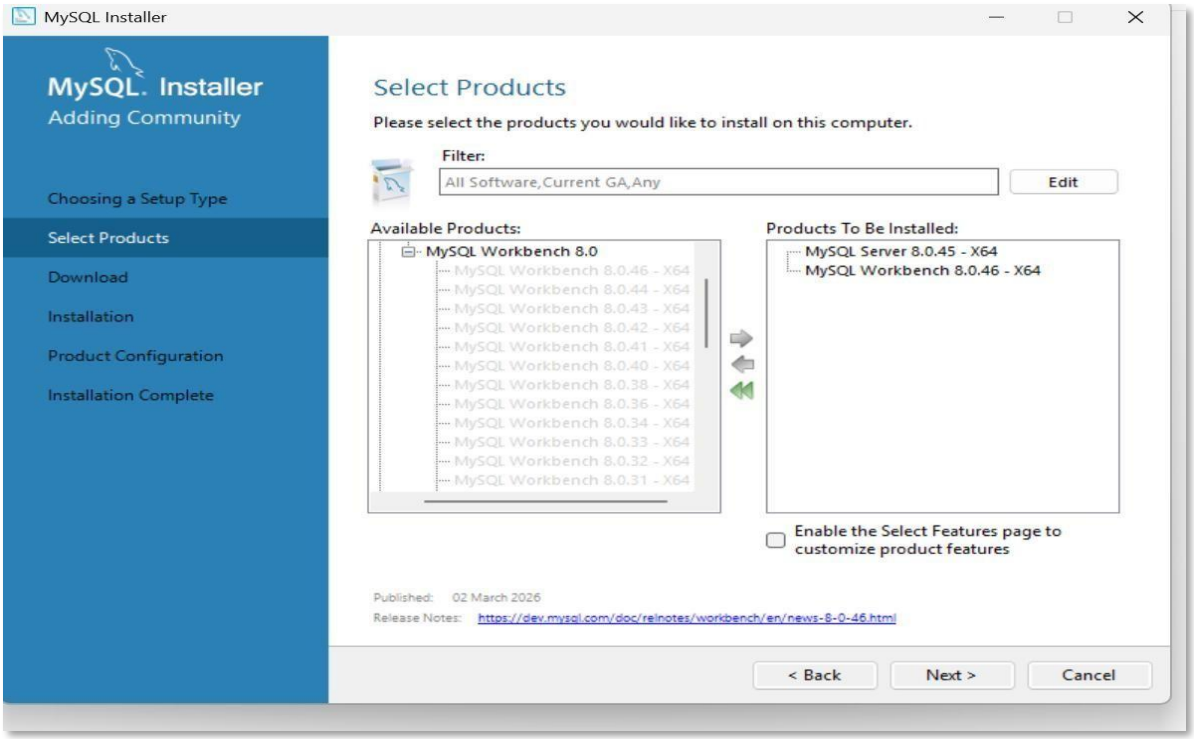
Task 4: On the Select Products page, click the + button next to MySQL Servers. Then, choose MySQL Server 8.0.45 – x64 from the MySQL Server 8.0 list.



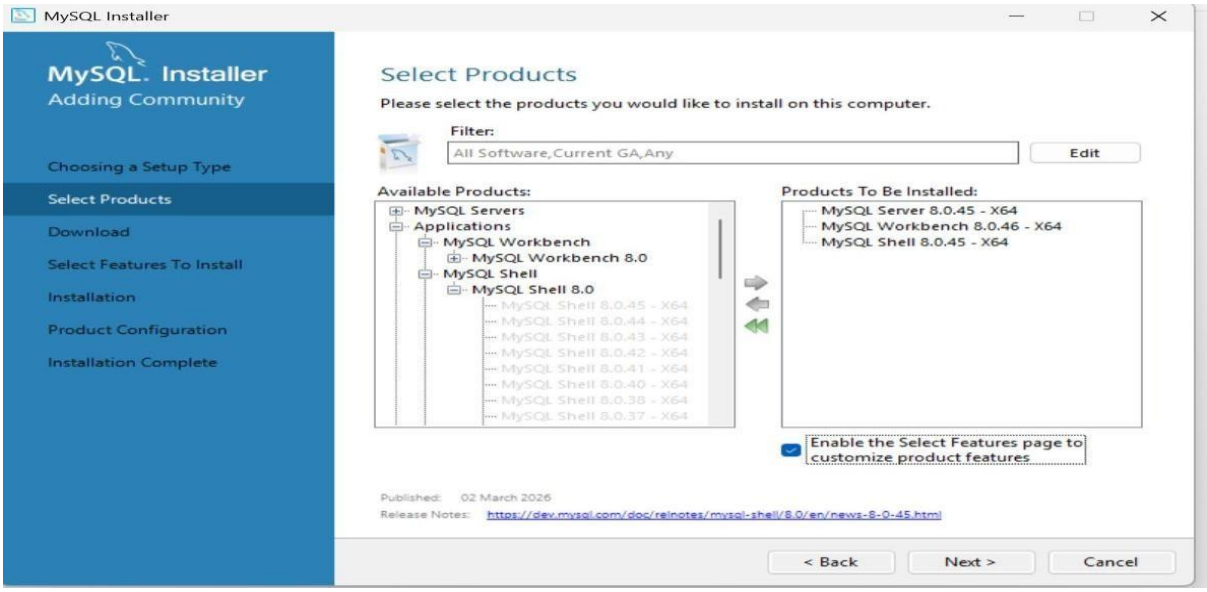
Task 5: Move your selected server to the Products to be Installed box.



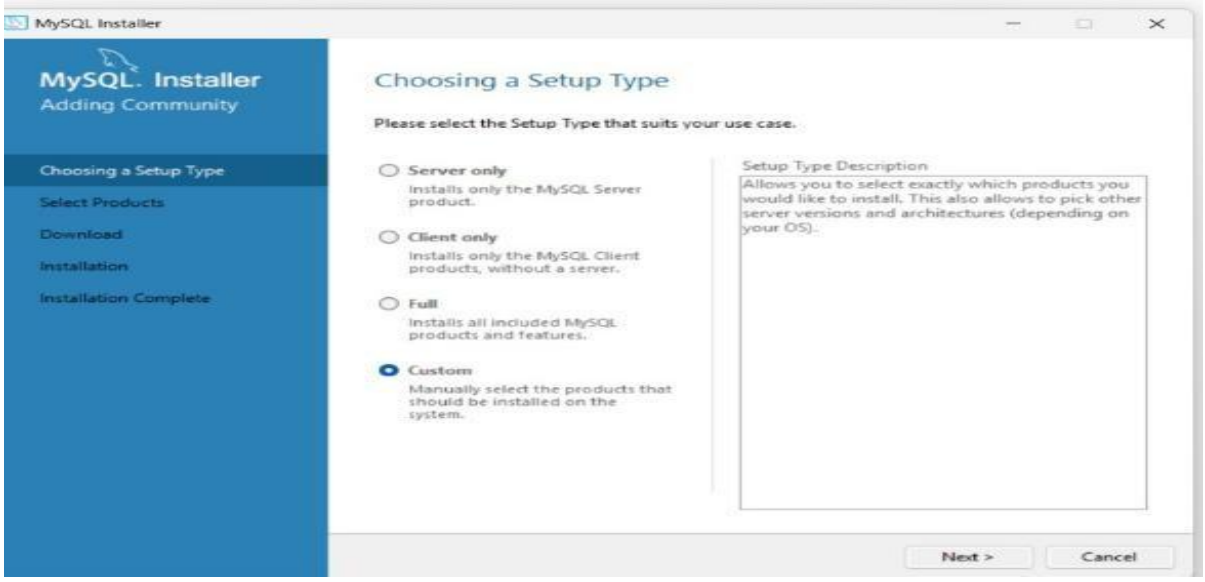
Task 6: Just like in Step 4, click the **+** button next to MySQL Workbench. Then select MySQL Workbench 8.0.46 – X64 from the MySQL Workbench 8.0 list and add it to the Products to be installed.



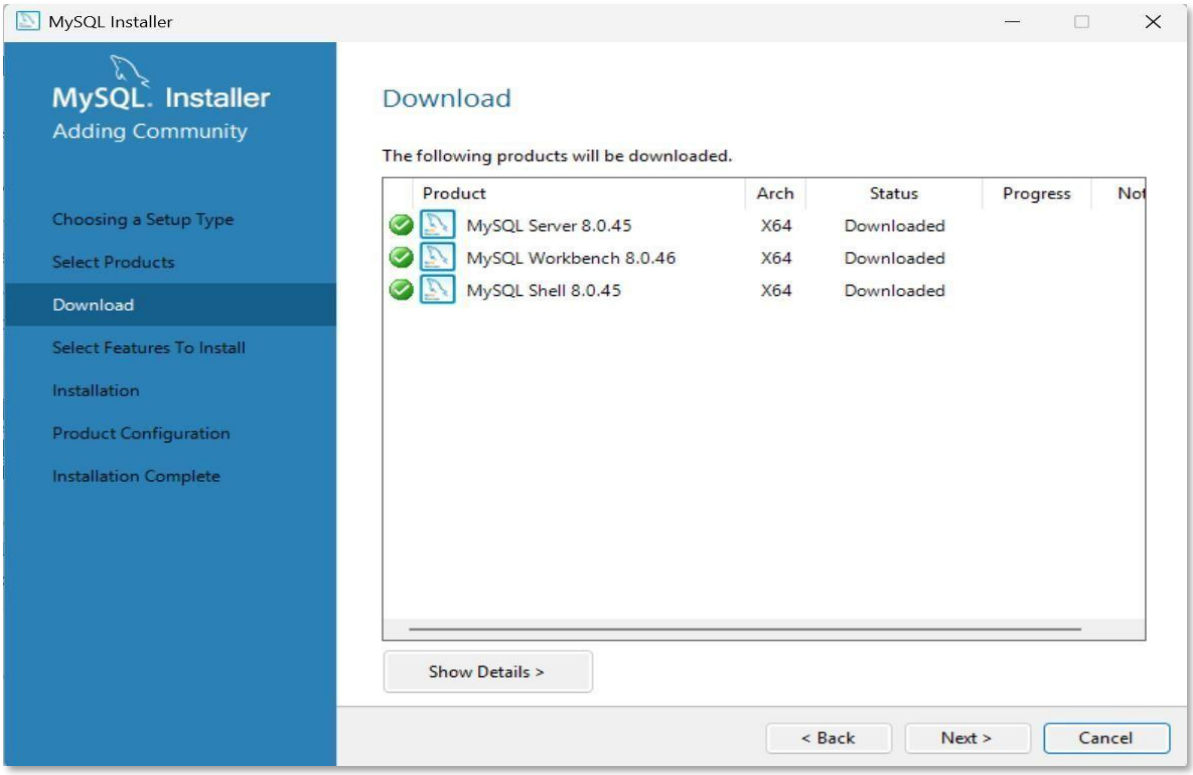
Task 7: Just like in Step 4 and Step 5, click the + button next to MySQL Shell. Then select MySQL Shell 8.0.45 – X64 from the MySQL Shell 8.0 list and add it to the Products to be installed.



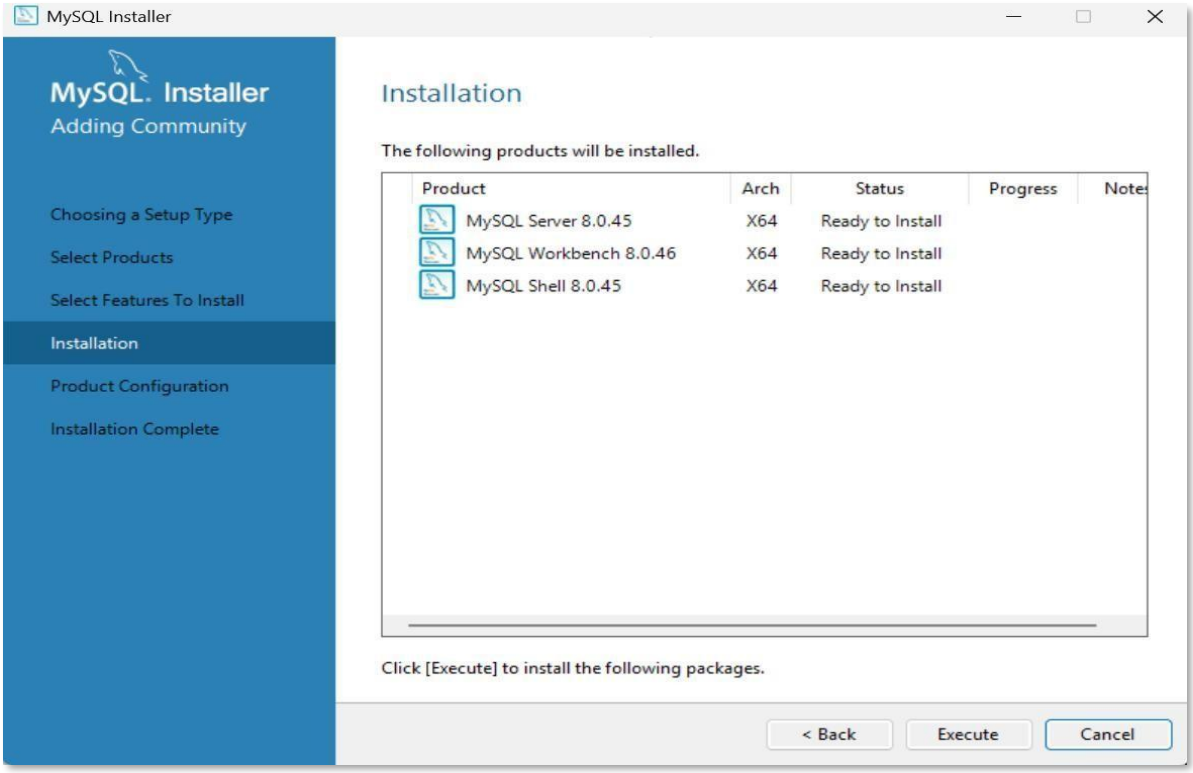
Task 8: Click Execute to start downloading and installing all the selected packages. Just sit back and let the installer do its work



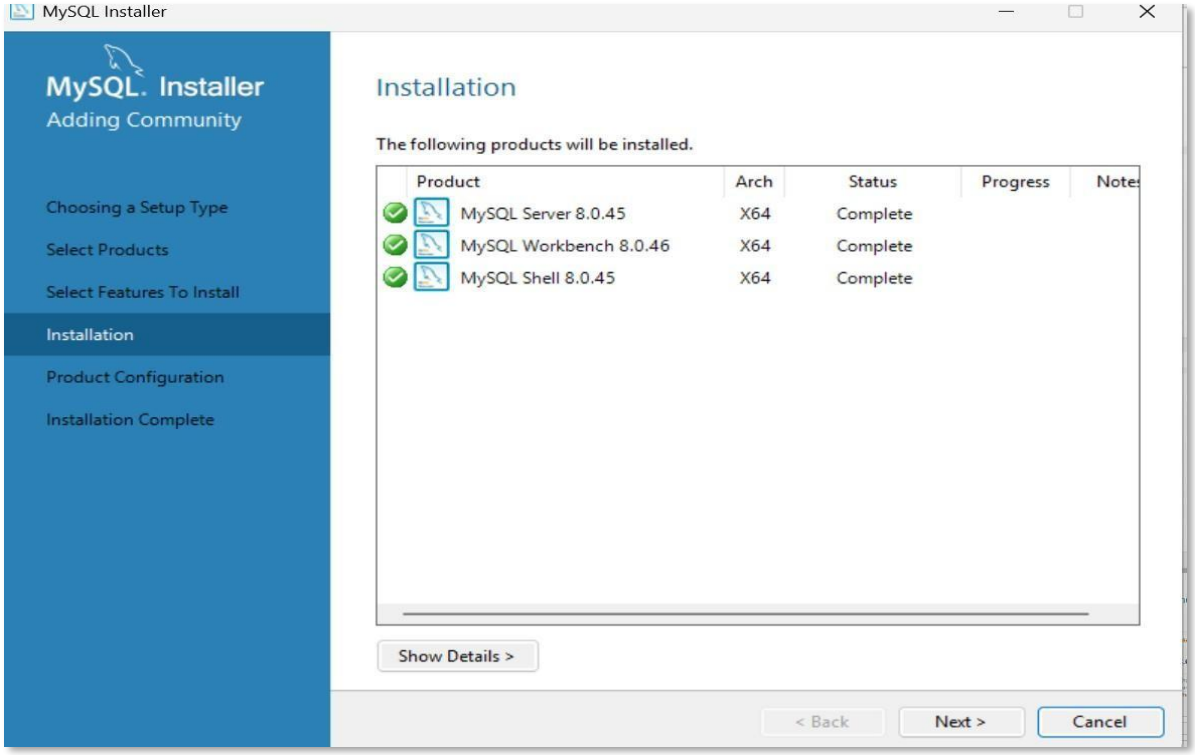
Task 9: After downloading your packages as shown in the picture below click on next.



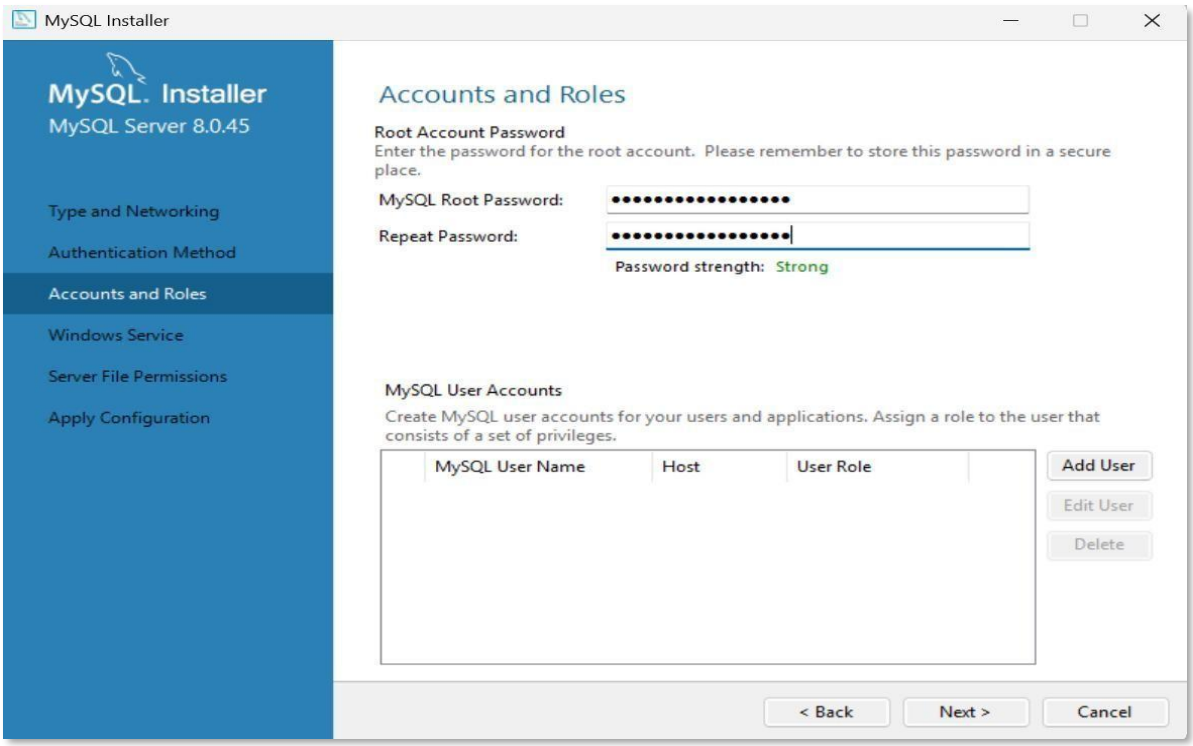
Task 10: Run your downloaded packages to start the installation. Just follow the prompts, and everything will be set up automatically.



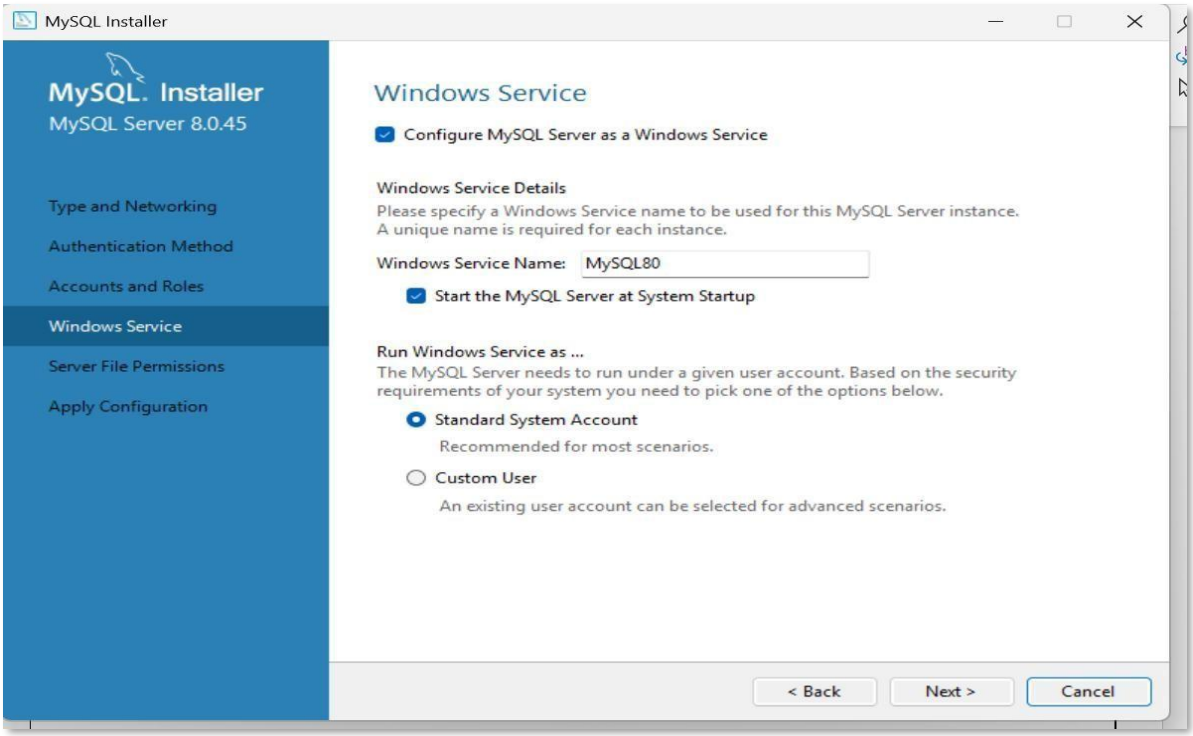
Task 11: Once your packages are installed (as shown in the picture below), click Next to continue.



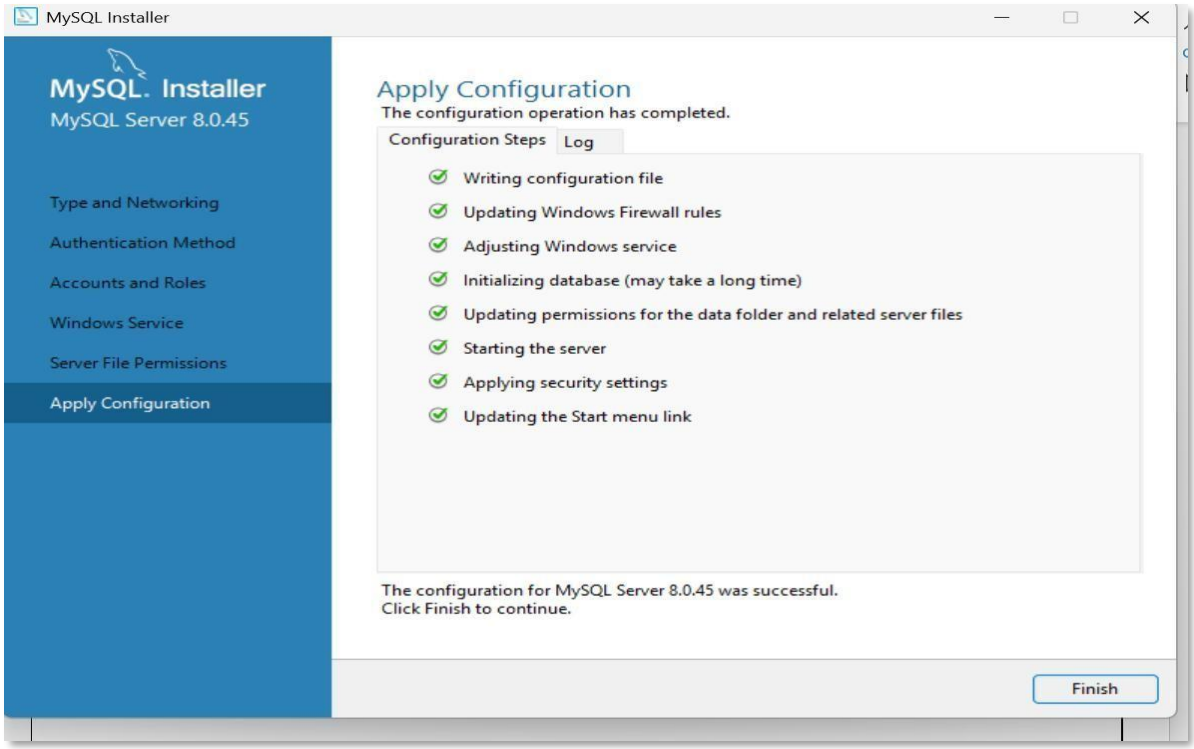
Task 12: Choose a strong password for your root account, then click Next to continue.



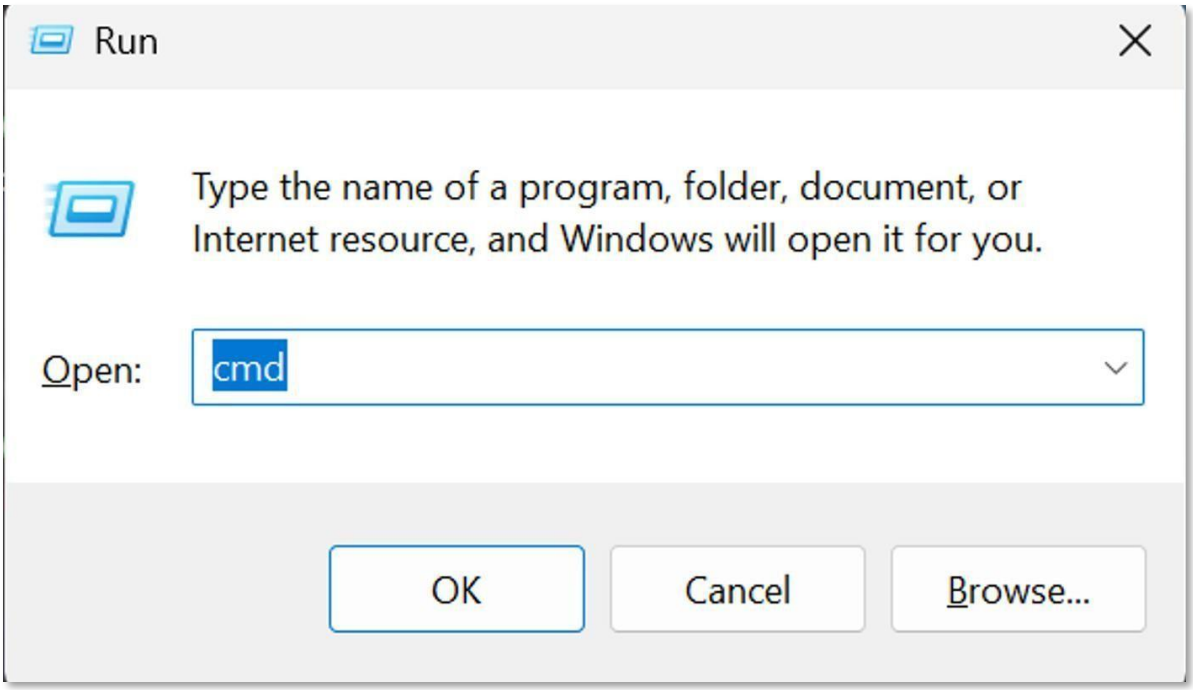
Task 13: On the Windows Service page, set up MySQL Server to run as a Windows service, then click Next to continue.



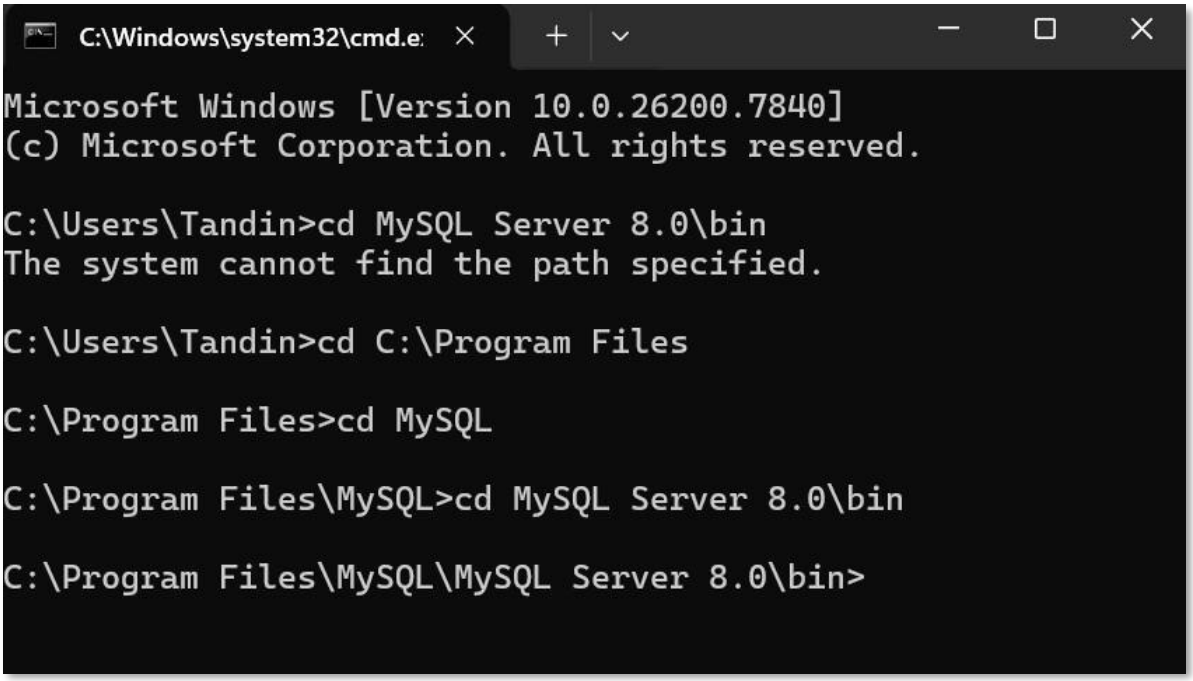
Task 14 Once the configuration is complete, click Finish to move on.



Task 15: Open your cmd file.



Task 16: Set your final directory to: C:\Program Files\MySQL\MySQL Server 8.0\bin



Task 17: Add MySQL to your system PATH, then open the MySQL Shell and type:

```
C:\Windows\system32\cmd.e:

Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Tandin>cd MySQL Server 8.0\bin
The system cannot find the path specified.

C:\Users\Tandin>cd C:\Program Files

C:\Program Files>cd MySQL

C:\Program Files\MySQL>cd MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.45 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show database;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'database' at line 1
mysql>
```

Conclusion

From this exercise, I learned the basic concepts of SQL and how to use it to create databases, manage tables, and retrieve data. The hands-on practice helped me better understand how relational databases work and how SQL commands are used in real situations. At first, I faced some difficulties with syntax errors and remembering the correct commands. However, through practice and experimenting with different queries, I was able to understand them better and improve my skills. This exercise helped me gain more confidence in working with databases and SQL.

Reference

Oracle Corporation. (n.d.). Oracle Database documentation. https://www.oracle.com/database/

MySQL. (n.d.). MySQL documentation. https://dev.mysql.com/doc/

Silberschatz, A., Korth, H. F., & Sudarshan, S. (2020). Database system concepts (7th ed.). McGraw-Hill Education.

Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems (7th ed.). Pearson.